

- 10-2 Find $Z_{EQ}(s)$ in Figure P10-2. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

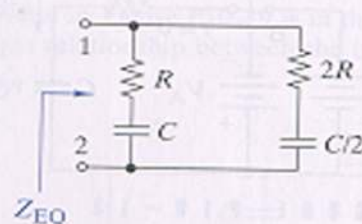


FIGURE P10-2

- 10-5 Find $Z_{EQ}(s)$ in Figure P10-5. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

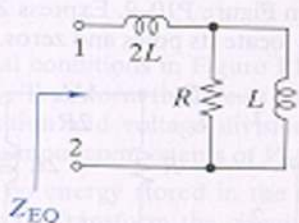


FIGURE P10-5

- 10-10 Find $Z_{EQ}(s)$ in Figure P10-10. Select values of R and L so that $Z_{EQ}(s)$ has a pole at $s = -1000$ rad/s. Locate the zeros of $Z_{EQ}(s)$ for your choice of R and L .

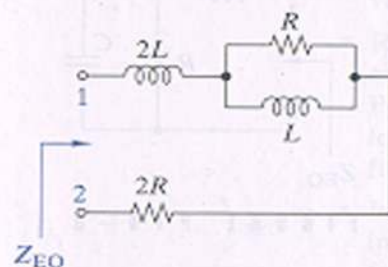


FIGURE P10-10

- 10-17 The switch in Figure P10-17 has been in position A for a long time and is moved to position B at $t = 0$.

- Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.
- Find $i_L(t)$ for $R_1 = R_2 = 500 \Omega$, $L = 250$ mH, $C = 4 \mu\text{F}$, and $V_A = 15$ V.

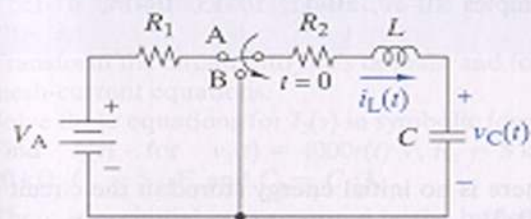


FIGURE P10-17

- 10-20 The circuit in Figure P10-20 is in the zero state. Find the s -domain relationship between the input $I_1(s)$ and the output $V_O(s)$.

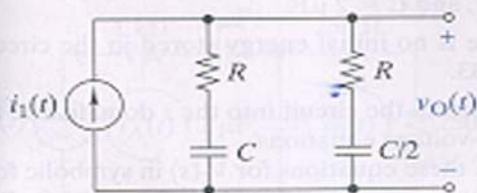
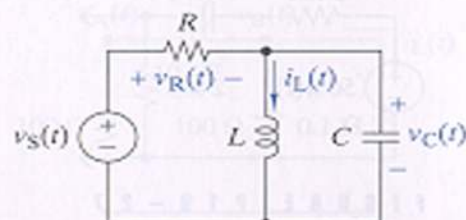
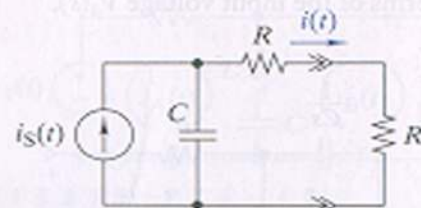


FIGURE P10-20

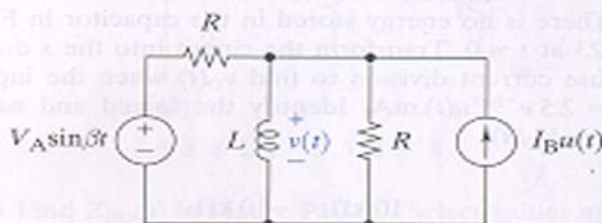
- 10-21 The initial conditions in Figure P10-21 are $v_C(0) = V_0$ and $i_L(0) = 0$. Transform the circuit into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_C(s)$.



- 10-25 The circuit in Figure P10-25 is in the zero state. Use a Thévenin equivalent to find the s -domain relationship between the input $I_S(s)$ and the interface current $I(s)$.



- 10-28 There is no initial energy stored in the circuit in Figure P10-28. Transform the circuit into the s domain and use superposition to find $V(s)$. Identify the forced and natural poles in $V(s)$.



- 10-31 There is no initial energy stored in the circuit in Figure P10-31.

- Transform the circuit into the s domain and formulate mesh-current equations.
- Solve these equations for $I_2(s)$ in symbolic form.
- Find $i_2(t)$ for $v_1(t) = 100u(t)$ V, $R_1 = 1$ k Ω , $R_2 = 2$ k Ω , $L = 4$ H, and $C = 500$ nF.

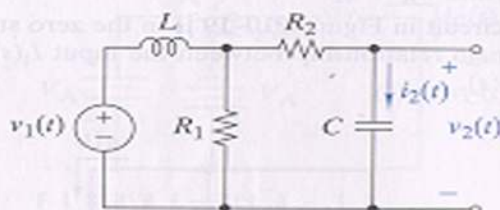


FIGURE P10-31

- 10-32 There is no initial energy stored in the circuit in Figure P10-32.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Solve these equations for $V_2(s)$ in symbolic form.
- Find $v_2(t)$ for $v_1(t) = 10u(t)$ V, $R_1 = R_2 = 500 \Omega$, $L = 0.5$ H, and $C = 2 \mu\text{F}$.